

MALEK BENCHEHIDA

(708) 691-1663 • benchehidam@yahoo.com

LinkedIn • github.com/malekbenchehida • malekbenchehida.github.io

EDUCATION

University of Illinois Chicago

Bachelor of Science in Computer Science

Chicago, IL

Expected December 2026

- **Leadership:** Technical Software Lead, SAE Formula Electric Team; led telemetry development for vehicle speed and throttle-pedal position (0–100%).

SOFTWARE ENGINEERING EXPERIENCE

MatchNice

Software Engineer

Chicago, IL (Remote)

June 2026 – Present

- Developed and launched a production donation-matching application alongside the CTO using Next.js, TypeScript, and Supabase PostgreSQL to replace a legacy no-code prototype.
- Enabled milestone-based fund matching in a recurring-donation prototype by building React campaign controls, Express APIs, PostgreSQL persistence, and an embeddable widget; contributed to vNext integration.
- Implemented webhook ingestion and sync workers for Fundraise Up and GoFundMe, checking provider event IDs before dispatch and relying on database uniqueness constraints to prevent duplicate records during concurrent retries.
- Reduced nonprofit research and sales-playbook turnaround from 18 hours to 15 minutes by automating URL intake, research, and document generation with Apps Script, Google Forms, and a Google Gem.

CAS Enterprise Inc.

DevOps & Full Stack Developer

Chicago, IL

Aug 2023 – June 2026

- Reduced client-side payload size by 30% by optimizing GraphQL queries and moving initial data fetching to the Next.js server.
- Cut Largest Contentful Paint from 3.4 s to 1.2 s and achieved 99/100 Lighthouse performance by combining server rendering, next/image hero preloading, and dynamic imports for non-critical JavaScript.
- Migrated application infrastructure from AWS to Azure, configuring App Service, Blob Storage, and Active Directory in the target environment.
- Reduced deployment turnaround from approximately 2 hours to 10 minutes by automating build and release workflows with Python.
- Resolved out-of-memory crashes by tracing oversized SQL results (exceeding 2M+ rows) with monitoring scripts, batching processing, and adding indexes.

PROJECTS

EdgeGuardNano — Edge Telemetry & Inference

2026

- Enabled offline telemetry processing on an NVIDIA Jetson AGX Thor by implementing asynchronous, non-blocking Rust ingestion pipelines using ownership-based memory safety.
- Optimized LightGBM inference for ARM64 CPUs through profiling; streamed edge-node metrics and DDoS classifications to a Next.js dashboard through GraphQL subscriptions.

P2P Crypto Wallet Prototype

2025

- Protected balance consistency during concurrent transfers in Go and Node.js using PostgreSQL ACID transactions and SELECT FOR UPDATE row locks; validated failure recovery with automated rollback tests.
- Implemented BIP39 key generation and TOTP multifactor authentication for wallet access and state changes.

Real-Time Gesture Inference Engine

2025

- Built a webcam-to-prediction pipeline with MediaPipe, a Python FastAPI endpoint, and a scikit-learn random forest, returning gesture classifications to a JavaScript rock-paper-scissors application.

RESEARCH EXPERIENCE

Chicago Tech Circle

Machine Learning Research Collaborator

Chicago, IL

Sept 2026 – Present

- Reviewing chest X-ray and visceral-fat literature to guide an early-stage Python/PyTorch study of type 2 diabetes risk prediction.

TECHNICAL SKILLS

- **Languages:** Rust, Go, Python, TypeScript, JavaScript, SQL, Java, C++, C
- **Databases & Infrastructure:** PostgreSQL, AWS, Azure, Docker, Linux, Supabase, MongoDB
- **Backend & APIs:** Node.js, Express.js, FastAPI, GraphQL, REST APIs, OpenAPI, Drizzle ORM, Zod
- **Delivery & Tooling:** CI/CD, GitHub Actions, Git, Apps Script
- **Frontend & ML:** React, Next.js, Tailwind CSS, scikit-learn, LightGBM, MediaPipe